

School Website – Product Design Document

2026-09-17 · @Someone

Built up role by role. Each section covers the client's raw requirement plus where automation removes manual work.

Correction (added after Teacher discussion began): this build is for a **Nepal** school. Earlier cost/legal/vendor references researched under an assumed India context — Indian payment gateways, the Indian DPDP Act, INR pricing — have been corrected where they appear below. Anything not yet flagged should be re-verified against Nepal-specific sources before being treated as final.

Mind map — all roles

Image — School website mind map, all roles

Each branch box's subtitle is the key confirmed decision for that area — full detail for each sits in the "Detailed data & role connections" section under its role heading below.

1. Student

Client requirement (raw)

- Self-signup form: first/middle/last name, DOB, phone, parent name + phone, previous school/college, joining year, school/college, course/class, year, admission reference, certificate upload, email, password.
- System auto-generates student ID (college/year/course/roll-no format).
- Submitted form goes pending until staff verification; student notified by SMS + email once verified; login enabled only after verification.
- Dashboard: personal details, course/class, certificate, fees (total/paid/remaining/history, online payment + voucher upload), attendance, results by exam term/year, notes & question papers by subject, homework/assignment submission.

- Student cannot edit own details. Shared notes/question papers can't be downloaded or screenshotted. **Student can download their own data** (fee receipts, results, personal details, certificates) — confirmed by client.

Automation opportunities

Stage	Automation	Effect
Signup	Cascading dropdowns (school → course → year)	Fewer wrong/mismatched selections
Signup	Auto-fill from admission reference (if tied to an enquiry record)	Student retypes nothing already on file
Signup	Real-time validation (phone format, DOB vs. course level, duplicate email/phone)	Errors caught before submission, no rejection cycle
Signup	Student ID shown immediately post-submit	Reference number available before approval completes
Verification	Auto-notify on every status change (received / under review / approved / needs correction)	No silent wait; zero staff effort per notification
Verification	Auto-route to correct approver by course/departement	No duplicate checks, nothing falls through
Fees	Remaining balance auto-computed the instant a payment is verified	Dashboard always accurate, no manual staff update
Fees	Reminders before due date and if overdue	Removes routine follow-up calls
Fees	Receipt auto-generated on payment verification	Nothing manual to issue
Results	Auto-notify the moment results are published	Student doesn't have to check manually
Attendance	Percentage auto-computed, flagged below threshold (e.g. 75%)	Proactive nudge, not a surprise at exam time

Stage	Automation	Effect
Notes/homework	Auto-tagged by subject, date, teacher on upload	No flat file list to hunt through

Open questions

- Should admission-reference auto-fill pull from an existing enquiry/CRM record, or is admission reference just a free-text field for now?
- Attendance alert threshold — confirm the percentage (75% used as placeholder above).
- Certificate upload: accepted file types and max size?

Detailed data & role connections (mind-map source)

Structured so each area can become a mind-map branch: node → fields → flow → connected roles → automation → decisions/open items.

Fees

- **Fields:** total course fee, paid so far, remaining balance, payment history (date/amount/method/status), due date(s)
- **Flow:** Accountant sets fee structure (per course/year) → Student sees amount due → pays **online** (gateway confirms instantly) or **offline** (uploads voucher → Pending review) → staff verifies offline payments → balance auto-updates → receipt auto-generated
- **Connected roles:** Accountant (creates structure, verifies offline payments) · Co-ordinator (can also verify, per spec) · Admin/Principal (views fee reports, exports)
- **Automation:** live balance computation, due-date reminders, auto receipt generation
- **Decided:** accountant adds fees, students pay online or offline — confirmed

- **Decided:** payment gateway — Nepal-specific, not Razorpay/PayU (correction: earlier open item assumed an Indian context). Real options are eSewa, Khalti, or Fonepay — Nepal's dominant wallets/gateways. Khalti specifically offers a ready-made "School Fee Payment" product built for Nepali schools/colleges (parents pay via Khalti wallet, bank/mobile banking, or card; SMS confirmation on payment), which could shortcut a custom integration. Alternative: a multi-gateway aggregator (e.g. PayBridgeNP, APINepal) that wires eSewa + Khalti + Fonepay through one API/dashboard if the client wants parents to have payment-method choice rather than being locked to one wallet. API access requires the school to be a registered institution in Nepal; typical setup cost per gateway is in the Rs 20,000–25,000 range (Fonepay/eSewa/Khalti), with some alternatives (ConnectIPS, IME Pay) offering free API access — plus per-transaction fees typically 1–4%. **Open:** partial payments, late fees, refunds, per-student fee customization — still unresolved, and which specific gateway/aggregator to commit to needs a client decision, not just a technical one.

Attendance

- **Fields:** overall %, subject-wise %, daily/session log, threshold alert
- **Flow:** Teacher marks attendance manually per session/subject → system aggregates into running % → student dashboard reflects near real-time → auto-alert if below threshold
- **Connected roles:** Teacher (marks it, sees own attendance too) · Admin/Principal (sets attendance timing/in-out window) · Parent (potential notification target, phone on file)
- **Automation:** live % computation, threshold alert, trend indicator
- **Decided:** display-only for student, no student-side action
- **Open:** exact threshold %, parent notification scope, correction/dispute flow, whether "in/out time" implies check-in/check-out tech (QR/geofence) or just a marking window

Results

- **Fields:** marks per subject, term/terminal, auto-computed grade, overall marksheet
- **Flow:** Teacher enters marks → Co-ordinator verifies all teachers' marks → grade auto-computed from co-ordinator's rules → Co-ordinator publishes → student notified → **Recheck:** student requests on a specific mark → Co-ordinator always notified, Principal notified above a threshold (TBD) → Co-ordinator can edit → grade/% auto-recalculates → student notified of any change
- **Connected roles:** Teacher (enters marks) · Co-ordinator (verifies, sets grade rules, publishes, handles recheck edits) · Principal (notified on large rechecks only)

- **Automation:** auto-notify on publish, auto-notify on any post-publish change, auto grade computation
- **Decided:** recheck button exists, dual notification (co-ordinator always + principal above threshold), co-ordinator edits and republishes, student always notified of changes, grade boundaries are co-ordinator-configurable (not fixed school-wide) — confirmed
- **Open:** exact Principal-notification threshold (fixed marks vs. % vs. grade-boundary cross) — deferred, co-ordinator's call for now

Notes & question papers

- **Fields:** subject, uploader (teacher), upload date, file, "new" flag
- **Flow:** Teacher selects year/class/subject → uploads → goes live immediately, no approval step → student browses by subject → view/download blocked, teacher can delete + re-upload to replace (no version history kept)
- **Connected roles:** Teacher (uploads, deletes/replaces) — no other role touches this data
- **Automation:** auto-tagging by subject/date/teacher, "new since last visit" flag
- **Decided:** immediate publish (no approval step), delete-and-replace (not versioned), standard file limits (PDF/DOC/DOCX/JPG/PNG, 20MB), "no download/screenshot" is soft-enforced only (disable right-click/print + watermark with student ID/name) — client to be told explicitly this isn't a hard guarantee — confirmed
- **Open:** none remaining — fully decided

Homework / assignments

- **Fields:** assignment (subject, class, deadline), submission file, submission status (`Not submitted` / `Submitted` / `Late` / `Reviewed`), marks + feedback
- **Flow:** Teacher creates assignment with deadline → student submits file → auto-flag `Late` if past deadline → Teacher reviews and gives marks → student sees marks/feedback → **Resubmission:** student requests → teacher approves → new file replaces old one
- **Connected roles:** Teacher (creates, reviews, grades, approves resubmission) — no other role involved
- **Automation:** due-date reminder, late-flag, submission-count tally for teacher
- **Decided:** teacher creates assignments with deadlines, review + marks visible to student, resubmission is request-then-approve (gated, not self-serve) — confirmed

- **Open:** none remaining — fully decided

Login & authentication

- **Fields:** email (unique, validated at signup), password, one-time system-generated password (editable on first login), OTP (for forgot-password flow)
- **Flow:** Student sets email at signup → system validates uniqueness in real time during form fill → on account creation, student is issued a one-time password → student can change it after first login → **forgot password:** student requests reset → OTP sent to registered email → student sets new password
- **Connected roles:** none directly — self-service, but ties back to the same verification/notification system used for registration status
- **Automation:** real-time email-uniqueness check during form fill (before submission, not after), OTP generation and delivery for password reset
- **Decided:** login is email + password, unique email enforced at signup, one-time password issued initially and editable by student, OTP-based forgot-password flow — confirmed

Cross-area connections

- **Verification/notification thread:** Fees (accountant/co-ordinator verify) → Results (co-ordinator verifies/publishes) → all use the same "student gets notified on status change" automation pattern — one notification system serves all three
- **Co-ordinator is the hub:** touches Fees (structure + verify), Results (verify, grade rules, publish, recheck edits), and student verification at signup — heaviest-loaded role in the student journey
- **Principal only enters the loop** for large recheck escalations — otherwise stays at the reporting/oversight layer, not day-to-day student data
- **Parent phone/email** (captured at signup) is a potential notification branch for Fees (reminders) and Attendance (threshold alerts) — not yet confirmed in scope

2. Teacher

Client requirement (raw)

- Teacher is created by Co-ordinator, Admin, or Principal — not self-signup (raw spec; superseded, see decision below: only Co-ordinator creates teachers).
- Creation form includes: subject, class, and time (per class/subject).

- Teacher selects year, class/course, subject to upload notes and question papers.
- Teacher can see student SID + name (filtered by year, course/class, subject) to enter marks.
- Teacher can see assignments/homework by year and class.
- Teacher can manually mark attendance per subject.
- Teacher can see their own attendance.
- Teacher can add a free-text daily activity note per subject/class (what was taught/done, dated).
- **Confirmed:** one teacher can teach multiple classes and different subjects.

Detailed data & role connections (mind-map source)

Account creation & login

- **Fields:** name, email (unique, validated), phone, subject(s) — multiple allowed, assigned class(es)/course(s) — multiple allowed, teaching schedule/timing, one-time password
- **Flow:** Co-ordinator/Admin/Principal fills teacher creation form → system generates login credentials → one-time password sent to teacher's email → teacher logs in with email + password → changes password on first login → forgot password uses OTP to registered email (same pattern as student)
- **Connected roles:** Co-ordinator only — Admin and Principal cannot create teacher accounts (deviates from raw spec, confirmed by client)
- **Automation:** auto-generated credentials + email delivery on creation, dashboard pre-filled with assigned classes/subjects on first login
- **Decided:** one teacher can be assigned multiple subjects and multiple classes — confirmed
- **Subject allocation pipeline:** Co-ordinator creates subjects (master list, per year/class/course) → Co-ordinator (primarily) assigns teacher to subject(s) + class(es) + time slot, selecting only from existing subjects — no inline subject creation during teacher assignment. teacher creation and subject/class assignment are both Co-ordinator-only — Admin/Principal have no role in creating or assigning teachers — confirmed

- **Marks entry**

- **Fields:** year, class/course, subject, exam terminal, student SID, student name, marks split into components (e.g. theory/practical), submission status
- **Flow:** Teacher selects year → class/course → subject (only from their own assigned subjects) → system pre-lists SID + name for that class → teacher enters component-wise marks per student → submits → status **Under review** → Co-ordinator verifies → locked (further changes go through the recheck flow)
- **Connected roles:** Co-ordinator (verifies, can trigger recheck edits) · Student (sees published result) · Principal (only on large recheck escalations, per Results flow)
- **Automation:** bulk-entry grid (spreadsheet-style, whole class at once), auto-flag students with missing marks before submission, draft-save for partial entry
- **Decided:** marks are split into components per subject (not a single number), teacher can freely edit marks after submitting up until Co-ordinator verifies — confirmed
- **Open:** Decided (cont.): components differ per subject — configurable, not the same theory/practical split for every subject — confirmed. Open: max marks per component — fixed school-wide or configurable per subject?

- **OCR photo-based mark entry (flagged — v2, not v1)**
 - **Concept:** teacher photographs a structured, printed mark sheet (fixed SID + marks-box layout) instead of typing into the grid; OCR extracts the handwritten numbers and pre-fills the digital grid; teacher reviews and corrects before submitting — never auto-submits
 - **Feasibility:** real, but handwritten-digit OCR is meaningfully less reliable than printed-text OCR — misreads are expected, and a paper template must stay fairly rigid for row-mapping to work (a plain notebook page won't work)
 - **Cost model** (Google Cloud Vision OCR, checked Sept 2026): first 1,000 images/month free, then \$1.50 per 1,000 images. Example for a ~500-student school, ~6 subjects, 3 terminals/year — worst case, one photo per class-subject-term submission concentrated around exam windows: roughly 300–1,000 images/month during peak months, likely staying within the free tier most months; a larger school (~2,000 students) doing the same could see \$50–150/year total, still low relative to build cost. Actual cost depends on submission volume, image quality retries, and whether OCR runs once or is retried per misread.
 - **Recommendation:** v2 feature, not v1 — the manual grid already removes most friction (whole class, one screen); OCR's accuracy risk on handwriting means a full manual review step is still required either way, which narrows the time actually saved. Revisit once usage patterns from v1 show real teacher pain around manual entry at scale.

- **Attendance marking**

- **Fields:** class/course, date, student SID, name, status (Present/Absent only), marked-by (class teacher), marked-on (timestamp)
- **Flow:** designated Class Teacher for that class marks attendance once per day (bulk grid, mark-all-present default, flip absentees) → status locked at end of day, editable only same-day → auto-aggregates into student's running attendance % immediately
- **Connected roles:** Co-ordinator (assigns which teacher is the Class Teacher for each class, alongside subject assignment) · Student (sees own % and threshold alerts) · Admin/Principal (sets attendance timing window, per spec)
- **Automation:** mark-all-present default (teacher only flips absentees), auto-reminder if a class's attendance isn't marked by end of day, duplicate-prevention (can't mark same class twice same day)
- **Decided:** attendance is once per day overall, not per subject/period — deviates from raw spec ("teacher marks attendance according to their subject"); resolved by introducing a **Class Teacher** designation — Co-ordinator assigns one teacher per class as Class Teacher, responsible for the daily mark. Statuses are Present/Absent only (no Late state). Editable by the marking teacher only on the same day; locked after — confirmed
- **Resolved: "in/out time" (Admin/Principal setting per class/course, from raw spec) is dropped entirely — not implemented. Attendance stays a simple once-a-day Present/Absent mark by the Class Teacher, which must be completed before the day ends — confirmed.**

- **Facial recognition attendance (flagged — v2/v3, not v1)**
 - **Concept:** camera-based face match against enrolled student photos marks attendance automatically, no teacher input needed
 - **Approaches:** (a) dedicated camera hardware at gate/classroom — real-time scan as students enter; (b) mobile-app camera capture — Class Teacher points phone/tablet at class once, app does group face-match against the roster (no extra hardware)
 - **Context:** correction — the original note referenced an Indian state government deployment (Karnataka), which does not transfer to Nepal. That context has been removed; Nepal-specific vendor precedent has not yet been researched.
 - **Cost estimate:** correction — the figures previously here were sourced from the Indian market (INR pricing, Indian vendors) and don't reliably transfer to Nepal. Needs fresh research against Nepal-specific hardware/SaaS vendors before any number is treated as reliable. Directionally, expect the same shape of cost (a one-time hardware cost, or a per-student/year SaaS fee) — just not these specific figures.

- **Legal (Nepal — corrected):** Nepal's relevant law is the Individual Privacy Act, 2018 (Privacy Act 2075) and Individual Privacy Regulation, 2020, grounded in Article 28 of the Constitution — not India's DPDP Act, which does not apply here.
 - Biometric data is explicitly classed as Sensitive Personal Information, held to a stricter consent standard than ordinary data
 - Consent must be informed and specify purpose, scope, recipients, and retention period — implied or pre-ticked consent is treated as insufficient for sensitive categories like biometric data
 - No codified, specific statutory requirement for parental/guardian consent for a minor's biometric data (unlike India's DPDP Act) — a genuine gap in Nepali law, not a safe harbor; obtaining explicit guardian consent anyway is the safer practice given the general informed-consent requirement and a child's limited capacity to consent
 - Nepal has no dedicated data protection regulator — enforcement runs through the District Court via the Privacy Act; less regulatory clarity, not less legal exposure (violations are a criminal offence under the Act)
 - Nepal's Ministry of Education has published sector-specific student-data-protection guidance for institutions — worth checking for anything more specific to schools before implementation
- **Recommendation:** v2/v3 feature, not v1 — not a technical blocker, but it adds a parental-consent workflow, a data-retention policy, and real regulatory exposure that nothing else in this build requires. The Class Teacher manual-mark system already locked in is free and has zero legal complexity — revisit facial recognition only if the client wants to pursue it explicitly, with legal counsel involved for the consent process.

- **Content upload (notes & question papers) — Teacher side**
 - **Fields:** year, class/course, subject (teacher's own assigned subjects only), file, upload date
 - **Flow:** teacher selects year → class/course → subject → uploads file → goes live immediately, no approval step, visible instantly to students in that class/subject. To replace: delete old file → upload new one (no versioning kept)
 - **Connected roles:** none — no Co-ordinator/Admin/Principal approval or visibility step
 - **Automation:** auto-tagged by subject/date/teacher, "new since last visit" flag on student side
 - **Decided (carried over from Student section):** immediate publish, delete-and-replace not versioned, standard file limits (PDF/DOC/DOCX/JPG/PNG, 20MB), "no download/screenshot" soft-enforced only (right-click/print disabled + watermark) — not a hard guarantee, client informed explicitly

Assignments — Teacher side

- **Fields:** subject, class (teacher's own assignments only), deadline, assignment description/instructions, optional teacher file attachment, student submission file, submission status (Not submitted/Submitted/Late/Reviewed), marks, feedback
- **Flow:** teacher creates assignment (subject/class, deadline, instructions, optional attached file) → students submit files → auto-flag Late if past deadline → teacher reviews, gives marks + feedback → visible to student → resubmission: student requests → teacher approves → new file replaces old
- **Connected roles:** none — purely Teacher ↔ Student
- **Automation:** auto-flag Late submissions, submission-count tally for teacher ("18 of 30 submitted"), due-date reminder to students
- **Decided:** teacher can optionally attach a file (e.g. instructions PDF) when creating an assignment — not text-only — confirmed

- **Daily activity log**
 - **Fields:** subject, class, date (auto-defaults to today), free-text entry, teacher (implicit)
 - **Flow:** teacher selects subject + class (own assignments only) → types what was taught/done that day → auto-dated → saved. Visible to students/parents on the student dashboard as a lesson log per subject.
 - **Connected roles:** Student/Parent (view only) · Co-ordinator/Principal (can reference for syllabus tracking, not spec'd as an approval step)
 - **Automation:** auto-dated entry (no manual date picker for the common case), quick-entry prompt surfaced right after attendance marking for that class (piggybacks on an existing daily habit rather than being a separate destination), mandatory-entry reminder if a teacher hasn't logged a class's activity by end of day
 - **Decided:** visible to students/parents (not internal-only), mandatory per class per day with a reminder if skipped — confirmed
 - **Note:** this adds a new item to the Student dashboard not originally listed in the raw spec's student dashboard bullet — worth reflecting there too (daily lesson log, browsable by subject, alongside notes/papers and homework)

3. Co-ordinator

Built area by area. All five Co-ordinator areas are done: Admissions and verification, Results and marks cards, Teacher attendance and search, Setup (classes and subjects), and Website content with two-factor login.

Client requirement (raw): Admissions and verification

- Co-ordinator registers a student by filling the registration form.
- Co-ordinator verifies registered students in a "verify student" tab.
- The raw spec also gives a verify-student tab to the Accountant and to Admin/Principal. Superseded by the decision below.

Client requirement (raw): Results and marks cards

- Check results by class, year and subject.
- Verify marks entered by all teachers.
- Create marks cards by selecting a template or design.
- Set grades according to marks.
- Publish results by year, class and subject; once published, students see them by year and terminal.
- See the result of an entire class on a single marks card by year, class and terminal.
- See the top 20 students of the whole college or school by year and terminal.

Client requirement (raw): Teacher attendance and search

- Co-ordinator marks teacher attendance.
- Co-ordinator can search any student by SID, phone number or name, by year, and see personal details, result, fees and attendance.
- The raw spec also gives the Co-ordinator creation of the fee structure by class and year. Superseded by the decision below.

Client requirement (raw): Setup

- Co-ordinator creates subjects for different year, class and course.
- The spec does not say who creates the classes and courses themselves.

Client requirement (raw): Website content and two-factor login

- Co-ordinator manages the website: changing or editing posts and other content such as notices, holidays, routines, vacancies and anything else that needs editing.
- Two-factor login for the Co-ordinator (also listed for Accountant and Admin/Principal, not for Super Admin).
- Admin/Principal can also edit and maintain the website; Super Admin manages the website and branding (logo, fonts, colours, school name).

Detailed data & role connections

Admissions & verification

- **Confirmed flow:** two ways in, one "Pending review" queue. Students sign up themselves, or the Co-ordinator registers a walk-in with the same form.

- **Confirmed review screen:** one applicant per card, details beside the certificate, three actions: Approve, Ask for changes (Co-ordinator picks fields and writes a reason; student edits only those), Reject (reason required). Status labels in plain English: Pending review, Needs your attention, Approved, Rejected.
- **Confirmed on approval:** system generates the student ID (college/year/course/roll no.) with the roll number assigned at approval, not signup, so rejected applicants leave no gaps. SMS and email go out and login switches on. A duplicate check flags a matching phone number, or name plus date of birth.
- **Decided:** only the Co-ordinator approves new students. Deviates from the raw spec, which also lets the Accountant and Admin/Principal verify students; flag for the client. Recommendation, not a rule: allow more than one person to hold the Co-ordinator role so approvals do not stall on leave.
- **Decided:** walk-in students registered by the Co-ordinator are auto-approved. The student ID is generated immediately and login details are sent by SMS and email. The audit log records who registered them.
- **Decided:** a rejection is final. The applicant must start a new application. A rejected application releases its email and phone number so they can reapply; the duplicate check still shows "previously rejected". Ask for changes is the fix-it path.
- **Decided:** admission reference is an optional free-text box labelled "Referred by (if any)" until the client clarifies what it means.

Results & marks cards

- **Confirmed flow:** a results inbox with one card per class, subject and terminal, with statuses Waiting for teacher, Ready to verify, Verified and Published. Verify shows the teacher's grid read-only with missing or unusual marks highlighted; actions are Approve or Send back with a note, plus bulk approve for a whole class. Builds on the decided Draft → Under review → Published pipeline and recheck routing.
- **Decided:** results are published for a whole class at once, per terminal. Publish stays disabled until every subject for that class is Verified, and the inbox shows which subject is holding it up. Deviates from the raw spec (publish by subject); flag for the client. Confirmed: automatic reminder to the teacher whose subject is blocking.
- **Decided:** each class or course has its own grading scale. Confirmed: a new class starts by copying another class's scale; a class with no scale cannot be published; a scale change affects only unpublished results.

- **Decided:** marks cards use 2–3 ready-made templates only, with the school logo, name and colours applied from Super Admin branding. The Principal's signature and school seal are uploaded once and appear on every card. The signature and seal go beyond the raw Super Admin spec; add them when Super Admin is designed.
- **Confirmed:** a whole-class view as one sheet of students × subjects with totals and rank, exportable.
- **Decided:** the Top 20 list is visible to students as well as staff. **Confirmed:** students see name and rank only (no ID or phone), for their own year and terminal, and only after that class's results are published; staff see the whole school by year and terminal.
- **Open question:** is the client comfortable showing rankings to other students, given many are minors? Confirm with the client.

Teacher attendance & search

- **Confirmed flow (attendance):** a daily list of all teachers, pre-filled as Present. The Co-ordinator changes exceptions to Absent or On leave and saves. No clock times, since in/out tracking is already dropped. Teachers see their own attendance as a read-only monthly calendar. A monthly count of present, absent and leave days per teacher is generated automatically.
- **Decided:** past days stay editable by the Co-ordinator, with a reason recorded in the audit log (who, when, why). This is looser than the student attendance rule (same-day edits only). **Confirmed:** the teacher gets a short notice when their past attendance is changed.
- **Confirmed search:** one search box in the staff console (name, SID or phone). A result opens the student's profile with tabs Personal, Results and Attendance, and a year picker that covers "according to year" in the spec.
- **Decided:** the Co-ordinator can correct a student's personal details, with a reason required; the audit log records who changed what and why. **Confirmed:** the student ID can never be edited, and the student gets a short notice when their details change.
- **Decided:** fees are Accountant-only. The Co-ordinator's student profile has no Fees tab, and fee structure creation (raw spec item j) moves to the Accountant. Deviates from the raw spec (item m lists fees for the Co-ordinator); flag for the client. The fee-structure part is an interpretation; confirm when Setup is discussed.

Setup: classes & subjects

- **Confirmed flow:** one Classes & subjects screen. The Co-ordinator picks a class or course, sees its subjects as a list and taps Add subject. Each subject has a name and its mark components (theory/practical split set per subject, as already decided). The class's grading scale (already decided per class or course) is set on the same screen.
- **Confirmed:** at the start of a new year each class copies last year's subjects and the Co-ordinator edits only what changed. A subject that already has marks or an assigned teacher can be archived but not deleted, so old results keep their subject.
- **Decided:** the Co-ordinator creates the classes and courses, then adds subjects to them. This fills a gap in the spec.
- **Decided:** each class has one designated Class Teacher, chosen by the Co-ordinator, who marks student attendance once a day. There is no backup teacher; the Co-ordinator can reassign at any time. Confirmed: a reminder to the Co-ordinator if a class has no attendance marked by end of day.
- **Decided elsewhere:** fee structure creation is with the Accountant, not in Setup.
- **Open question:** do college courses run by year or by semester? The spec only says year. Ask the client.

Website content & two-factor login

- **Confirmed flow:** one Website section with five content types: Notice, Holiday, Routine, Vacancy and Post. Each is a short form with a title, text, optional image or file, a publish date and an optional hide-after date. Expired items hide automatically, and every item can be previewed before it goes live. Routines come from the timetable (teachers' class and subject times) so nobody types them twice. Vacancies show a contact email or phone; a job-application system is out of scope for v1. Branding stays with Super Admin.
- **Decided:** the Co-ordinator drafts website content and the Principal approves before it goes live (Draft → Waiting for approval → Live). Any holder of the Admin role (Principal, Director, trustees) can approve; Admin and Principal are one role. Confirmed: the Principal's own edits go live directly; an Urgent flag notifies the Principal immediately; the Admin role is held by several people, so approvals do not stall.
- **Decided:** staff two-factor login is by SMS code. Confirmed: also apply it to Super Admin (the raw spec omits it); codes expire after a few minutes; "remember this device for 30 days"; "send by email instead" if the SMS does not arrive. Needs an SMS service, which student verification messages already require.

- **Resolved:** Admin and Principal are one role, held by several top people (Principal, Director, trustees). Any holder can approve. See section 5.
- **Open question:** Nepal SMS providers and cost per message need verifying.

Open questions

- What is "Admission reference" meant to capture? The spec (item k) does not say. It could be a referral, an entrance-test number or an agent code. Ask the client.

4. Accountant

Built area by area. All four Accountant areas are done: Fee structure, Admission and discount, Payments and verification, and Search.

Client requirement (raw)

- Create the fee structure by year, class or course.
- Admit a student by filling the registration form.
- After selecting the student's year and course, show the fee structure and total fees for the course. The Accountant can enter the actual admission fee manually or give a discount; after a discount, total fees and actual fees are the same.
- Search by SID, phone number or name to see complete student details: personal details, result, attendance and fees.
- A verify area with two tabs, verify student and verify fees, showing student details and the fee voucher, where the Accountant can mark verified.
- Add a payment for a student.
- Two-factor login.
- Admin/Principal item: fees can be monthly, yearly or course-wise.
- Already superseded: the verify-student tab (only the Co-ordinator verifies students). See Co-ordinator, Admissions & verification.

Detailed data & role connections

Fee structure

- **Proposed flow:** one Fee structure screen. The Accountant picks a class or course and a year, then adds fee items such as Admission fee, Tuition and Exam fee. Each item has an amount and a billing type: One-time, Monthly, Yearly or Whole course. The spec names monthly, yearly and course-wise; one-time is an addition for items like the admission fee. The total is calculated automatically, in NPR. Each item has a due date, or a day of the month for monthly items. The student dashboard shows one "next due" card. A new year starts from a copy of last year's structure and the Accountant edits only what changed. Late fees and fines are out of scope for v1.
- **Decided:** fees are fixed for one academic year, and the same structure applies to every student in a class. The Accountant can change fees only when setting up the next year's structure. Discounts stay per student (see Admission and discount).
- **Decided:** the Principal approves each yearly fee structure before it goes live (Draft → Waiting for approval → Live). Proposed: any holder of the Admin role, as with website content; a class and year with no live structure cannot take admissions; a locked structure can be unlocked by the Principal with a reason recorded.
- **Decided elsewhere:** fee structure creation moves from the Co-ordinator to the Accountant, and the Co-ordinator has no Fees view. Staff two-factor login is by SMS.
- **Open question:** what does "course wise" mean? A single payment for the whole course, or a course total paid in instalments? Ask the client.
- **Open question:** does the school use Bikram Sambat (BS) dates for the academic year and due dates, or AD? Ask the client.

Admission and discount

- **Proposed flow:** the Accountant registers a student with the same registration form. The student goes to the Co-ordinator's Pending review queue, because only the Co-ordinator approves students; auto-approval applies only to the Co-ordinator's own walk-ins.
- **Proposed fee screen:** picking a year and course shows the fee structure and total. The payable amount starts equal to the total and cannot be typed over freely. Any reduction is a discount, so the raw spec's "enter actual admission fee manually" becomes a discount.

- **Decided:** the Principal approves every discount. The Accountant enters an amount or a percentage and a reason (Scholarship, Sibling, Staff child, or Other with a note). Status runs Waiting for approval → Approved or Declined. Until approved, the student's payable stays at the full fee. Deviates from the raw spec, where the Accountant gives discounts directly; flag for the client.
- **Proposed:** after approval, total fees and actual fees are the same (as the spec says), the original total is shown struck through, and the student's dashboard shows a Discount line. A declined discount leaves the full fee in place and the Accountant can propose another.
- **Proposed:** the first payment is recorded only after the Co-ordinator approves the student, so a rejected applicant never has money to refund.
- **Proposed:** one Approvals inbox for Admins covering website content, yearly fee structures, discounts, payment reversals and refunds, each with Approve and Decline.
- **Open question:** which discount reasons does the client actually use? Confirm the list with the client.

Payments and verification

- **Proposed flow:** a payment arrives in one of three ways. (1) Online through the gateway (eSewa, Khalti or Fonepay): the gateway confirms it, so it is marked paid automatically with no voucher to verify. (2) Bank or QR deposit with a voucher upload: it appears in the Accountant's verify queue as Pending review. (3) Cash at the counter: the Accountant taps Add payment (amount, method, date) and it is recorded at once.
- **Proposed verify screen:** one card per voucher showing the student, the amount claimed and the voucher image. Actions are Verify or Reject with a reason. A rejection shows the student "Needs your attention" with the reason. A reused reference number gets a duplicate warning. The verify area is fees only, since only the Co-ordinator verifies students.
- **Proposed:** a verified payment is taken off the remaining balance (as the spec says) and creates a numbered receipt the student can download. Partial payments are allowed and applied to the oldest due item first.
- **Proposed:** a payment can never be deleted. A mistake is reversed, and the audit log keeps who, when and why.
- **Decided:** the Principal approves every payment reversal. The wrong entry stays, marked "Reversal pending", and the balance changes only when the Principal approves. Proposed: reversals join the Admin's Approvals inbox, which then covers website content, yearly fee structures, discounts, reversals and refunds.

- **Open question:** which gateway does the client want (eSewa, Khalti or Fonepay), and does the school already have a merchant account with one? Ask the client.

Search

- **Proposed flow:** the same single search box as the Co-ordinator's (name, SID or phone). The student's profile opens on the Fees tab first, showing the total, any discount, paid, remaining, upcoming dues and payment history with receipts. Quick actions: Add payment, and Propose discount if none is set.
- **Decided:** the Accountant's student profile has two tabs, Personal and Fees only. Result and Attendance are hidden. Deviates from the raw spec, which lists personal details, result, attendance and fees; flag for the client. Reason: fee work does not need academic records, and hiding them protects student privacy.
- **Proposed:** the Personal tab is read-only for the Accountant; corrections stay with the Co-ordinator.

5. Admin (Admin/Principal)

Built area by area. All Admin areas are done: Approvals inbox, Reports and exports, Accounts and search, and Audit trail.

Client requirement (raw)

- Register a student.
- Create teacher, co-ordinator and accountant.
- See the total number of students and teachers.
- See students by class or year with filters, with Excel export.
- See total, paid and remaining fees of the school and college, with year and class filters and Excel export.
- Search any student or teacher by ID, phone number or name.
- Set attendance timing by class or course (in and out time).
- Fees can be monthly, yearly or course-wise.
- See the whole school or college result on one sheet, class-wise or year-wise.
- See which action was performed by which user (Co-ordinator, Accountant or self).
- Verify students in a verify tab.
- Two-factor login.

- Edit and maintain the website.
- Already superseded: create teacher and verify student (Co-ordinator only), attendance in/out timing (dropped), and fee billing types (covered by the fee structure).

Detailed data & role connections

Role definition

- **Decided:** Admin and Principal are one role, held by several top people (Principal, Director, trustees). Every holder has the same powers. The spec's seven roles become six: Super Admin, Admin, Co-ordinator, Accountant, Teacher, Student. Deviates from the raw spec, which lists Admin and Principal separately; flag for the client. Wherever this doc says "Principal approves", it means any Admin.

Approvals inbox

- **Proposed flow:** one inbox as the Admin's home screen, urgent items first, with a badge count. Five kinds of item: website content, yearly fee structures, discounts, payment reversals and refunds. Each card shows a type label, who asked and a one-line summary; tapping it opens the detail (website preview, fee structure table, discount with reason, or the original payment for a reversal). Buttons are Approve and Decline; declining needs a reason that goes back to the requester, who can resubmit. A History tab lists approved and declined items. Admins get a reminder if an item is still pending after two days; urgent website items notify them immediately.
- **Decided:** any one Admin can approve an item; there is no second approver. Whoever acts first resolves it and it disappears from every other Admin's inbox. The audit log shows who approved it.

Reports and exports

- **Proposed flow:** an Overview screen with four tappable numbers (Students, Teachers, Fees collected, Fees remaining), each opening its report. Every report uses the same two optional filter chips (Year and Class) and one Export to Excel button.
- **Proposed students report:** counts by class and year; tapping a class lists its students.
- **Decided:** the Fees report shows total, paid and remaining, plus total discounts given, and a student-wise dues list (who owes what), with the same filters and an Excel export. Goes beyond the raw spec, which asks for totals only.

- **Proposed results report:** a summary per class (pass rate, average, top score) that opens into the class marks sheet (see Co-ordinator, Results). The Excel export holds the full sheet. Only published results appear. Replaces the raw spec's single sheet for the whole school, which would be too large to read.
- **Open question:** the spec says "school and college". If both run in one system, a School / College filter chip is needed. Ask the client.

Accounts and search

- **Proposed team screen:** a list of staff (Co-ordinators and Accountants) with an active or inactive status. Add person asks for a role (Co-ordinator or Accountant), name, phone and email. The system generates the login and sends it by SMS and email, as for teachers, and the person sets up two-factor login on first sign-in.
- **Proposed:** an Admin deactivates an account rather than deleting it, so history in the audit log stays intact, and can reactivate it later. There can be more than one Co-ordinator or Accountant. New Admin accounts are created only by Super Admin, as the raw spec says.
- **Proposed search:** one search box covers students and teachers, each result labelled by type. A student result opens four read-only tabs (Personal, Results, Attendance, Fees). A teacher result shows details, subjects and attendance, read-only. Corrections stay with the Co-ordinator.
- **Decided:** Admin does not register students; the Co-ordinator and Accountant do. Deviates from the raw spec (item a); flag for the client.

Audit trail

- **Proposed flow:** one Activity screen, newest first. Each row is a plain sentence naming who (name and role) did what, with the time; tapping it shows the detail (before and after, and the reason where one was required). Filters: person, role, type of action (Students, Fees, Results, Attendance, Website, Accounts), date range, and a search by student name or SID. An Export to Excel button, as on the other reports.
- **Proposed:** it records every action already promised in this doc, including approvals and declines, walk-in registrations, attendance edits, detail corrections, payments, discounts, reversals, result verification and publishing, website publishing and account changes. Admins' own actions are recorded too (the spec's "or self").
- **Proposed:** nobody can edit or delete an entry, including Super Admin. Only Admins and Super Admin can view it.

- **Decided:** the audit trail also records sign-ins, including failed two-factor attempts, in a separate Sign-ins tab so the main Activity screen stays clean.

Website and two-factor login

- **Decided elsewhere:** Admin holders edit and maintain the website. Their own edits go live directly, and they approve the Co-ordinator's drafts (see Co-ordinator, Website content & two-factor login). Staff two-factor login is by SMS code.

6. Super Admin

All Super Admin areas are done: Role definition, Branding, and Accounts and safeguards.

Client requirement (raw)

- Create Admin/Principal accounts.
- See all details and manage all users.
- Manage the website.
- Upload the school or college logo.
- Change font style, text colour and background colour.
- Add the school or college name that shows in the navbar and everywhere else it is needed.

Detailed data & role connections

Role definition

- **Decided:** Super Admin is the master role held by the build team, not a school role. School staff never see it; from the school's side there are five roles (Admin, Co-ordinator, Accountant, Teacher, Student). It is used for emergencies, support and major edits, and has access to everything. The raw spec reads as if Super Admin were a school role; flag for the client.
- **Proposed:** every Super Admin action is written to the audit log and shown to the school's Admins in the Activity screen with a "Support" label. Super Admin access is used only for recovery and support. The client agrees to this access up front, because students include minors; worth checking against Nepal's privacy law.

Branding

- **Decided:** only Super Admin manages branding: logo, school or college name (shown in the navbar and wherever else it is needed), font style, text colour and background colour, plus the Principal's signature and school seal used on marks cards and receipts. Changes to any of these go through the build team.
- **Decided:** free choice of font, text colour and background colour, with a readability check that warns or blocks when the contrast is too low. Proposed: a live preview before saving.

Accounts and safeguards

- **Decided:** Super Admin creates Admin accounts, one per person (so several for the Principal, Director and trustees), as in the raw spec. Proposed: Super Admin can see and manage all accounts, and resets access when someone loses their phone and cannot pass the SMS two-factor step. Super Admin has the same website tools as Admin, and its own edits go live directly. Two-factor login applies to Super Admin.
- **Open question:** does the client understand that branding changes and new Admin accounts go through the build team? Confirm with the client.

7. Data lifecycle & cross-role rules

How the student record lives across years, and the rules that cut across roles. Added after the end-to-end data walkthrough.

Student profile and years

- **Decided:** the student is the golden record. The profile (SID, personal details, parent details, certificates) carries forward to the next year automatically. Each year's records (fees ledger, attendance, marks and results, submissions, receipts) belong to that year. No edits are allowed to a previous year's data.
- **Proposed:** the SID is permanent and never changes. The class roll number belongs to each year's record. Personal details are shared across years, but a published marks card keeps the details as they were that year.
- **Proposed:** year rollover is one Co-ordinator screen per class. Each student is marked Promote, Repeat or Leaving, then Confirm. It needs the new year's fee structure to be live (Admin-approved). On confirm, new-year records are created and the old year locks.

- **Proposed:** an unpaid balance from a locked year appears as "Previous dues" in the new year. Payments toward it are new entries in the current year, and the old year is never edited. Reversals and refunds work the same way: new entries that point to the old one.

Leave case: a student leaves, transfers or graduates

- **Proposed:** three statuses: Active, Left (transfer or dropout) and Graduated. The Coordinator marks a student as Left or Graduated with a reason and an effective date.
- **Proposed:** from the effective date no new fees are generated, and the student drops out of class lists, attendance grids, teacher marks grids and future Top 20 lists. Admin reports count Active students separately from Left and Graduated.
- **Proposed:** the student keeps read-only access to their own records (results, receipts, certificates), as decided for own data. Nothing is ever deleted.
- **Proposed:** the Accountant sees the student's dues at the moment of leaving. Dues stay on record.
- **Proposed:** a returning student is reactivated on the same profile and SID, with a new year record. No second profile is created.
- **Decided:** a student can be marked Left or Graduated only when their dues are zero. Dues means fees billed and unpaid up to the effective date; future fee items stop at that date. The Mark as Left button shows the dues and stays disabled until they are cleared. In the year rollover screen, a student marked Leaving must also have zero dues before Confirm. Proposed: dues the school agrees to forgive are cleared through the existing discount flow (Waive dues, Admin approves), so no new approval type is needed.

Refunds

- **Decided:** a refund is initiated by the Accountant and approved by Admin (any holder of the Admin role).
- **Proposed:** a refund is different from a reversal. A reversal corrects a wrong entry; a refund returns real money for a valid payment, for example after a student leaves or an overpayment. It goes to the Admin's Approvals inbox as a fifth kind of item.
- **Proposed:** the Accountant picks the student and payment, enters the amount (never more than the net paid) and a reason. After Admin approves, the Accountant records how the money was returned (cash, bank or wallet) with a reference. The ledger gets a negative refund entry and a refund receipt. The original payment is never edited.

- **Open question:** what are the client's rules on when a refund is allowed? Ask the client.

Parents

- **Decided:** parents and the student share one profile and one login. There is no separate parent account.
- **Proposed:** SMS notifications (fees due, results published, absence) also go to the parent phone number on the profile.